Emil I. Jaffal

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Education

City University of New York, The Graduate Center — Expected 2028
Ph.D., Chemistry
Advisor: Dr. Anton Oliynyk

Fordham University, Fordham College at Rose Hill — May 2023
B.Sc., Chemistry

Research Experience

Ph.D. Student — *New York, NY*
City University of New York, Hunter College — Jul 2024 – Present

Solid-State Laboratory – Dr. Anton Oliynyk

- Conducting exploratory syntheses of novel intermetallic materials with corresponding analyses using powder X-ray diffraction and scanning electron microscopy.
- Enhancing machine learning applications to predict properties of various binary and ternary compounds, focusing on improving interpretability and predictive capabilities of models in solid-state materials by incorporating detailed structural information.
- Directly mentoring a handful of students, providing guidance in their research projects with both experimental and computational techniques.

Undergraduate Researcher — *Bronx, NY*
Fordham University — Sep 2021 – May 2023

Organic/Materials Laboratory – Dr. Julia Schneider

- Steered materials research involving various reactions as part of a novel multi-step synthesis to create organic semiconductors (OSCs) with tunable conjugated heterocycles to improve conductivity.
- Instrumentation experience includes handling UV-Vis, NMR, fluorescence, and IR spectroscopy with respective machinery and analytic interpretations. General synthesis and purification skills include distillations, extractions, filtrations, and recrystallizations.

Computational Laboratory – Dr. Joshua Schrier

- Identified probable transition states of novel syntheses as part of an NSF-funded collaboration within the chemistry department.
- Performed numerous Gaussian ab initio calculations of internal energies, electronic structures, and geometric data using density functional theory to analyze reaction thermodynamics and predict isomer formations of OSCs.

Professional Experience

Research Chemist — *Tarrytown, NY*
ICL Industrial Products — Sep 2023 – Jul 2024

- Developed novel flame retardant (FR) blends for polyurethane foams in collaboration with external manufacturers and customers, ensuring compliance with international safety regulations.
- Pioneered the integration of polyurethane for battery encapsulation, contributing to advancements in sustainable and high-performance materials, leading to a patent application.
- Led application efforts for VeriQuel® F series, a proprietary phosphorus-based FR for flexible foams,

conducting iterative testing with customers and scaling toll production for MT quantities, with early sales reaching $800K.

- Executed laboratory experiments, standardized flammability tests, and physical property assessments to support new product development and market-driven innovations in halogenated and non-halogenated FRs.

Publications

**Indicates corresponding author*

15. Systematic Ternary Exploration and Expansion of the Ternary Dy–Ir–Sn System with Magnetic Investigation. *In preparation.* Apr 2026
Jaffal E.I., Brodska A., Selvaratnam B., & Oliynyk A.O.

14. Systematic Ternary Exploration and Expansion of the Ternary Gd–Os–Ge System. *In preparation.* Apr 2026
Jaffal E.I., Oziel J., Selvaratnam B., & Oliynyk A.O.

13. Achieving a Scalable Machine Learning Workflow for Crystal Structure Discovery with Experimental Validations. *Preprint submitted, Digit. Discov.* Apr 2026
Shiryaev D., **Jaffal E.I.**, Lee S., Selvaratnam B., & Oliynyk A.O.

12. Evaluation of Machine Learning Models in the Search of Interpretable Structure Maps of Perovskites and Spinels through Modified SISSO Routine. *Preprint submitted, Solid State Commun.* Mar 2026
Selvaratnam B., Oziel J., **Jaffal E.I.**, & Oliynyk A.O.

11. Disentangling structural patterns in quaternary germanides with materials informatics and exploratory synthesis: RE_4IrInGe$_4$ (RE = Y, Ce–Nd, Sm, Gd, Ho–Er) series. *Accepted, Inorg. Chem.* Mar 2026
Ponomarev A., Selvaratnam B., **Jaffal E.I.**, Shiryaev D., & Oliynyk A.O.

10. Improving Thermal Cycling Stability and Magnetocaloric Efficiency through Orbital Competition and Entropy Engineering in RE_{11}Rh$_4$In$_9$, 4*d*-4*f* Layered Intermetallics. *Chem. Mater.* Feb 2026
Tyvanchuk Y., ..., **Jaffal E.I.**, Selvaratnam B., Mudring A., & Oliynyk A.O.

9. Exploring Feature Engineering for Crystal Structure Classification: Interactive Applications of PCA & PLS-DA. *J. Chem. Educ.* Jan 2026
Shiryaev D., **Jaffal E.I.**, Selvaratnam B., Sun Y. & Oliynyk A.O.

8. RE_7M$_4$InGe$_{12}$ (RE = Y, Gd–Er; M = Rh, Os) Quaternary Germanide Series Expansion with Structural Motif Clustering. *Inorg. Chem.* Dec 2025
Pozdnyakova N., **Jaffal E.I.**, ... & Oliynyk A.O.

7. Investigating Mechanical Properties Through Defect Chemistry in Hard Binary Phosphide Material Ta$_3$P. *Solid State Commun.* Nov 2025
Jaffal E.I.,* Shiryaev D., Selvaratnam B. & Oliynyk A.O.

6. Dataset of Prototype Structures Adopted by Intermetallic Compounds with AB Stacking. *Data Brief.* Oct 2025
Selvaratnam B., **Jaffal E.I.**, Shiryaev D. & Oliynyk A.O.

5. Explainable Recommendation Engines to Predict Complex Intermetallics: Synthesis & Characterization of Gd$_{10}$RuCd$_3$, a Neutron Absorption Material. *J. Am. Chem. Soc.* Sep 2025

Xhabrahimi B., **Jaffal E.I.**, ... & Oliynyk A.O.

4. Materials Informatics Tools to Analyze Crystal Structures: Crystal Structure of Novel Ternary Indide $ErCo_2In$. *Integr. Mater. Manuf. Innov.* Jun 2025
Tyvanchuk Y., Lee S., ..., **Jaffal E.I.**, Selvaratnam B. & Oliynyk A.O.

3. Unsupervised ML Prediction of Novel 1:3 Intermetallic with Synthesis of $TbIr_3$ ($PuNi_3$-type) as Experimental Validation. *J. Am. Chem. Soc.* Feb 2025
Sethi S.S., Dutta A., **Jaffal E.I.**, ... & Oliynyk A.O.

2. Composition and Structure Analyzer/Featurizer for Explainable ML Models to Predict Solid State Structures. *Digit. Discov.* Jan 2025
Jaffal E.I., ... & Oliynyk A.O.

1. Synthesis of PyrDi Isomers w/ Tunable Excimer Formation. *Org. Lett.* Jan 2025
Johnston K., McCostis A., Mikita E., **Jaffal E.** & Schneider J.A.

Guest Lectures

Solid-State Chemistry: *Introduction to Thermoelectrics – Hunter College, NY* May 2025

Courses Taught

General Chemistry Lab (10600) – *Hunter College, NY* Aug 2025 – Present

Presentations

Brookhaven Lab Nuclear Chemistry Summer School – *New York, NY* Jul 2025
Using explainable recommendation engines for the discovery of $Gd_{10}RuCd_3$

ACS Mid-Atlantic Regional Meeting – *South Orange, NJ* May 2025
Quaternary Intermetallic Germanides: Structure, Properties, and Potential Applications

Fordham University Jean Dreyfus Lectureship – *Bronx, NY* Apr 2023
The Schneider Lab

Brookhaven Lab Nuclear Chemistry Summer School – *New York, NY* Jul 2024
The Oliynyk Lab

MAPS: Research at Fordham – *Bronx, NY* Nov 2022
Vinyl Azide Cyclization: Where Organic and Computational Chemistry Meet

Posters

North American Solid State Chemistry Conference – *Ames, IA* Jul 2025
Quaternary Intermetallic Germanides: Structure, Properties, and Potential Applications

Hackathons

SSMC-Collaboration Incubator – *Madison, WI* May 2025

Selected participant for national hackathon-style research workshop. Collaborated on the *Rational Design of Thermoelectrics* with an interdisciplinary cohort of PhD students and professors.

Selected Projects

See the full suite of apps on the Oliynyk lab website, both developed myself here.

XRD Comparison & Matching Tools Mar 2025

Contributed to the development and deployment of GUIs, allowing novice users to compare multiple '.xy' files from X-ray diffraction (XRD) data with .cif files. These help identify impurities, match phases, and visualize differences in crystal structure XRD patterns with ease.

Composition Analyzer/Featurizer (CAF) Jun 2024

Developed an interactive Python script that generates chemical compositional features and provides tools for filtering, sorting, and merging data. Aids novice solid-state/materials chemists in generating compositional training data for ML models ranging from dozens to tens of thousands of compounds.

Patents

Gadolinium–Ruthenium–Cadmium Intermetallic Compound with Negative Thermal Expansion and Neutron-Absorbing Uses Thereof. No. 63/997,273. Mar 2026

Emil Jaffal, Anton Oliynyk & Balaranjan Selvaratnam. *Patent pending.*

Heat Resistant Semi-Rigid Polyurethane Foams. *WO2026035457A1*. Feb 2026

Emil Jaffal, Sergei Levchik, Zhihao Chen, Jeffrey Stowell & Munjal Patel.

Awards and Grants

CUNY Conference Presentation Support Fund Apr 2026
Selected for $300 in reimbursement for conference presentation expenses.

CUNY Provost's Pre-Dissertation Science Research Fellowship Mar 2026
Awarded $4,000 to support early-stage summer research for the puprposes of dissertation proposal development.

CUNY Doctoral Student Research Grant (DSRG) Mar 2026
Awarded $1,239 to support travel for training in uranium intermetallic synthesis, as part of dissertation research and professional development.

CUNY Black and Schwartz Science Conference Travel Award Mar 2026
Awarded a prestigious $1,400 grant to attend a conference to present research.

NY-Hub Regional I-Corps Virtual Innovation Lean Bootcamp Jan 2026
Selected for the Winter 2026 cohort with access to a $3,000 customer discovery reimbursement fund for entrepreneurial development and startup experience.

Hunter College Cliff Sol Award Dec 2025
Awarded $500 for excellence by a graduate student.

CUNY Graduate Center Certificate of Achievement Jan 2026
Awarded $1,000 in recognition of outstanding success as a first-year student, including various publications during my first year.

CUNY Graduate Center Certificate of Achievement Jan 2025
Awarded $3,000 in recognition of the acceptance of a first-author publication during the Fall 2024 semester.

CUNY Graduate Center Science Scholarship Aug 2024

Fordham University Dean's List Nov 2023

NSF Summer Research Funding Grant (DMR-1928882) May 2022

Service

Content Creator of Educational Content Jun 2025 – Present
Founder/manager of the Oliynyk Lab YouTube channel for educational outreach.

Materials Today Physics – *Reviewer* Aug 2025 – Present

Fordham University Muslim Students Association – *Treasurer* Sep 2022 – May 2023

Fordham University Arabic Club – *Vice President* Jan 2022 – Aug 2022

Fordham Undergraduate Research Journal – *Peer Editor* Sep 2022 – May 2023

Students Mentored

Raoul Guillet (B.Sc. Chemistry – Fall 2025 - Spring 2026)
Anna Pomoraev (B.Sc. Chemistry – Spring 2025 - Fall 2025)
Alexandra Brodska (B.Sc. Biology – Fall 2025)
Brook Xhabrahimi (B.Sc. Chemistry – Spring 2024 - Spring 2025)
Natalia Poznyakova (B.Sc. Chemistry – Spring 2024 - Spring 2025)
Miriam Ismail (B.Sc. Chemistry – Spring 2025 - Spring 2026)
Riya Upadhyay (B.A. Human Biology – Spring 2025)
Alex Vtorov (B.Sc. Chemistry – Spring 2024 - Spring 2025)
Joseph Oziel (The Bronx High School of Science – Spring 2025 - Summer 2025)
Yujing Sun (The Bronx High School of Science – Spring 2025 - Fall 2025)
Brandon Lin (The Bronx High School of Science – Fall 2025)

Memberships

Sigma Xi, The Scientific Research Honor Society – *Associate Member* Mar 2023 – Present

Technical skills

Software: Adobe Illustrator, Bluehill, ChemOffice, Gaussian16, Mathematica, Maestro, Microsoft Suite, Signals Notebook, TopSpin, VASP, WebMO.
Programming & markup languages: Python, Bash, HTML, Wolfram (Mathematica).
Packages: NumPy, SciPy, Scikit-Learn, Pandas, Matplotlib.
Languages: English (native), Arabic (native), Spanish (conversational).